Solar Rooftop Feasibility Analysis Project

Ben Brown, Summer 2025

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1. Introduction

This project investigates the technical and financial feasibility of installing a residential solar photovoltaic (PV) system on my family's home. By integrating geospatial tools (QGIS), CAD modeling (AutoCAD), and performance simulation (PVWatts), I aimed to develop a detailed understanding of rooftop solar potential, system sizing, cost-benefit analysis, and environmental benefits. The project also served as a practical application of sustainable engineering concepts while developing proficiency in industry-standard software.

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2. Objectives

The primary objectives of this project were:

- To apply spatial analysis and 3D modeling tools to assess rooftop solar potential.
- To estimate solar panel system efficiency and annual energy output.
- To evaluate financial viability through cost, payback, and return on investment (ROI) analysis.
- To assess the environmental impact of switching to solar power.

To support this work, I completed the Coursera course "Solar Energy Basics" offered by The State University of New York, which provided a strong foundation for energy estimation and solar system planning.

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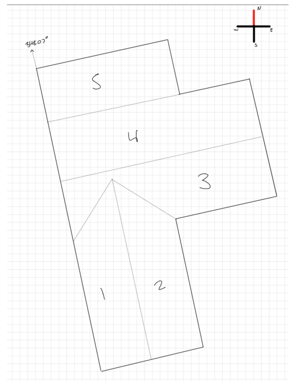
3. Methodology

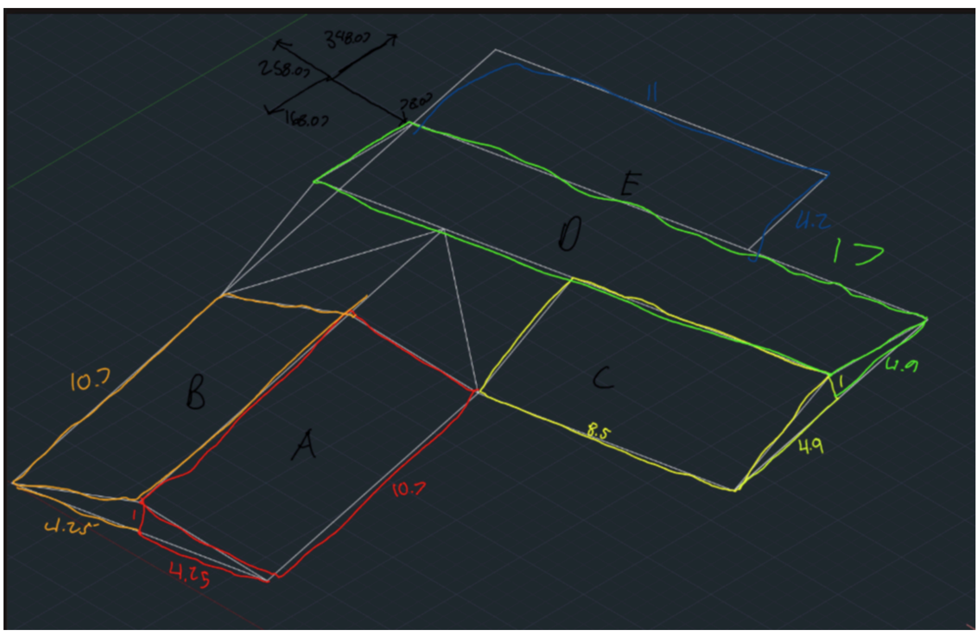
3.1 Roof Mapping Using QGIS

I acquired foundational QGIS skills from the YouTube tutorial "QGIS 3 for Absolute Beginners" by Klaus Karlsson. Using these skills, I identified the orientation, tilt, and dimensions of various roof sections by analyzing satellite imagery and generating accurate rooftop measurements.



3.2 3D Modeling Using AutoCAD

To construct a 3D model of my rooftop, I learned AutoCAD from the freeCodeCamp.org tutorial "AutoCAD for Beginners - Full University Course." This enabled me to create a digital twin of the roof and calculate usable surface area for solar panels, accounting for obstructions and angles.



3.3 Solar Efficiency Estimation

Using PVWatts and NREL data, I estimated an average irradiance of 1300 kWh/kW/year for my location. A formula was used in Google Sheets to calculate efficiency relative to the optimal tilt (21°) and azimuth (180°):

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=IF(TILT=0, 0.93, 1 - 0.0013*(TILT - 21)^2 - 0.0005*(ABS(AZIMUTH - 180))^1.35)
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With an assumed panel density of 0.17 kW/m² and surface area from QGIS/AutoCAD, I calculated system size and expected output.

Section Name	Area (m^2)	Tilt (°)	Azimuth (°)	Efficiency (%)	Panel Density (kWh/m^2)	System Size (kW/m^2)	Annual Output (kWh/kW)	Estimated kWh/year
A	46.72	13.24	78.07	66.46%	0.17	7.94	1300	6,861.79
B	46.72	13.24	258.07	74.23%	0.17	7.94	1300	7,664.51
C	42.5	11.53	168.07	86.92%	0.17	7.23	1300	8,164.06
D	85	11.53	348.07	37.83%	0.17	14.45	1300	7,106.55
E	46.2	21	180	100.00%	0.17	7.85	1300	10,210.20
Sum	267.14					45.41		40,007.10

3.4 Financial Analysis

Using an installation cost of $3,390/kW (average for Hawaii), I estimated the total system cost per roof section. After applying Hawaii's 35% solar tax credit, I calculated the net installation cost. By comparing solar output to household energy usage (5,639 kWh/year) and estimating revenue from surplus energy sold back to the grid, I determined the annual return and calculated the payback period.

	System Size (kW)	Installed Cost per kW ($)	Total Cost ($)	Tax Credit (35%)	Net Cost ($)	Annual Output (kWh/year)	Self-consumed (kWh)	Exported (kWh)	Electricity Rate ($/kWh)	Export Rate ($/kWh)	Bill Savings ($)	Export Credit ($)	Total Annual Return ($)	Payback Period (Years)
A	7.94	3,390.00	$26,924.74	$9,423.66	$17,501.08	6,861.79	5,639.00	1,222.79	$0.43	$0.14	$2,396.58	$171.19	$2,567.77	6.82
B	7.94	3,390.00	$26,924.74	$9,423.66	$17,501.08	7,664.51	5,639.00	2,025.51	$0.43	$0.14	$2,396.58	$283.57	$2,680.15	6.53
C	7.23	3,390.00	$24,492.75	$8,572.46	$15,920.29	8,164.06	5,639.00	2,525.06	$0.43	$0.14	$2,396.58	$353.51	$2,750.08	5.79
D	14.45	3,390.00	$48,985.50	$17,144.93	$31,840.58	7,106.55	5,639.00	1,467.55	$0.43	$0.14	$2,396.58	$205.46	$2,602.03	12.24
E	7.85	3,390.00	$26,625.06	$9,318.77	$17,306.29	10,210.20	5,639.00	4,571.20	$0.43	$0.14	$2,396.58	$639.97	$3,036.54	5.70
All	45.41	3,390.00	$153,952.78	$53,883.47	$100,069.31	40,007.10	5,639.00	34,368.10	$0.43	$0.14	$2,396.58	$4,811.53	$7,208.11	13.88

3.5 Environmental Impact Assessment

Using CencePower's kWh to CO_2 calculator, I quantified the environmental benefit of switching to solar, finding an estimated annual reduction of 2,071 kg of CO_2 emissions.

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4. Results

Rooftop mapping and modeling identified approximately 267 square meters of usable surface area for solar panels. This could support a system capable of generating more than 40,000 kWh/year, far exceeding the household's annual consumption. The system would result in significant energy surplus, enabling grid export and future capacity expansion (e.g., EV charging, air conditioning).

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5. Discussion

While technically feasible, a full-system installation would cost approximately $154,000 and have a payback period of over 13 years. This level of investment is currently beyond our household's budget. However, analyzing only section E of the rooftop revealed a more cost-effective solution. This section alone can generate sufficient electricity for our entire yearly consumption, with an installation cost under $26,000 and a payback period of just under 6 years. This modular approach enhances feasibility and provides a scalable path forward.

Assumptions include consistent panel performance, uniform irradiance, and static energy prices. Future refinements could involve shadow analysis, seasonal variation modeling, and contractor quotes.

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6. Conclusion

Following discussions with my family, the results suggest a promising opportunity for residential solar adoption, especially through phased implementation. This project allowed me to explore the intersection of spatial analysis, energy systems, and environmental impact. I developed technical skills in QGIS, AutoCAD, PVWatts, and financial modeling with Google Sheets. Most importantly, it deepened my commitment to sustainable engineering and renewable energy solutions.

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7. References

Karlsson, K. (n.d.). *QGIS 3 for Absolute Beginners* [YouTube video]. YouTube. https://www.youtube.com/watch?v=LfA3UgNHRt4 freeCodeCamp.org. (n.d.). *AutoCAD for Beginners - Full University Course* [YouTube video]. YouTube. https://www.youtube.com/watch?v=KAJpQ-s_U3o

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